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# Vinay Ram Gazula

vinay-ram1999.github.io
github.com/vinay-ram1999
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I am a passionate Mechanical Engineer and Data Science enthusiast eager to advance my career as a Data Scientist/Analyst. I am a well-organized and highly driven guy who always performs to the best of his capabilities. I contribute to open-source projects and I am open to work on projects involving Data Science, AI/ML, FinTech and CFD.

## Skills

| | | |
|---|---|---|
| **Programming** | Python*, R*, SQL, Bash*, C/C++, Matlab | |
| **Data Science** | Spreadsheets, Pandas, Tidyverse, Matplotlib, Plotly, Tableau | |
| **Tools** | **Git** | GitHub* |
| | **IDE** | KDE Advanced Text Editor (KATE)*, RStudio*, VS Code, VIM |
| | **Documentation** | MS Office*, LaTeX*, Markdown |
| | **Operating Systems** | Windows, Linux* |
| **Libraries** | NumPy, SymPy, yfinance, Tensorflow, ggplot | |
| **CAD & CFD Softwares** | Solidworks*, AutoCAD, CATIA, Tecplot 360*, Gmsh*, ANSYS Fluent | |

* Frequently used

## Projects

**MURPH Analytics** **Mar 2023 - Ongoing**
*Personal Project* *Project Source*

- Built an interactive dashboard using Streamlit to post-process and perform different analytical studies on the simulation data generated from the MURPH Solver.
- Implemented unstructured mesh plotting and 2D contour plots using Plotly.

**MURPH** **Dec 2022 - Ongoing**
*Personal Project* *Project Source*

- A Multi-dimensional (2-D/3-D), Unstructured-mesh and Reactions based Parallel solver for Hyper-sonic flow regimes.
- New design uproach is used to build the solver code so that the parallel code generation can be fully automated.
- 2D Ideal gas solver code is developed and it is in validation phase.

**AlgoTrade API** **May 2022 - Oct 2022**
*Personal Project* *Project Source*

- Fully automated NSE Stock/Equity trading-bot for Kotak Securities (KOTAK TRADE API) with integrated back testing and ML algorithms for price and trend predictions.
- The bot gathers any desired stock's historical price data, performs technical analysis using technical indicators and places orders to the stock broker using the broker API.

**Accelerating a Hypersonic $CO_2$ Reaction Solver** **Jul 2020 - May 2021**
*B.Tech Final Year Project* *Academic Project*

- A 2-D unstructured finite volume method based solver capable of simulating hypersonic flows in $CO_2$ rich Martian atmosphere with chemical kinetics for eight species consisting of $N_2, O_2, NO, O, N, CO_2, CO$ and $C$ incorporated in it is developed using python.
- The solver is parallelized to run on single or multiple CPUs and Nvidia GPUs in order to attain greater computational speeds using OP2 application programming interface.

**Design and Analysis of Heat Sink** **Oct 2020 - Dec 2020**
*Thermal Design of Electronic Equipment Course* *Coursework Project*

- Designed and analysed a 2-D heat sink made of Aluminium and a layer of epoxy between the chip and heat sink with specified thermal properties as a thermal design solution for cooling a noval chip (CPU for example) using CFD-HT (Computational Fluid Dynamics and Heat Transfer) software ANSYS Fluent.

## Professional Certifications

**Google Data Analytics** **Dec 2022 - Apr 2023**
*Coursera* *Certificate*

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## Publications

**Accelerating a Hypersonic $CO_2$ Reaction Solver** **Dec 2021**
*ISHMT-ASTFE Heat and Mass Transfer Conference* *Publication*

- Vinay, G., Dinesh, D., Satya, P., "Accelerating a Hypersonic $CO_2$ Reaction Solver", ***Proceedings of the 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference December 17-20, 2021, IIT Madras, Chennai-600036, Tamil Nadu, India.***

## Experience

**CAD Designer** **Mar 2022 - Apr 2023**
*Jayram Buildeng Solutions Pvt. Ltd.* *Vijayawada*

- Designing 2D CAD drawings using AutoCAD.
- Producing high quality, accurate, and timely reports for clients.

**Research Intern** **Sep 2021 - Feb 2022**
*Under Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP)* *SRM University AP*

- Worked on a Science Education and Research Board (SERB) govt. of India project titled "Wall Effects in Shock Wave Boundary Layer Interaction".
- Contibuted in the development of solver code for gaseous reactions and viscous properties.
- Validated the solver for different flow configurations like Single ramp, Double ramp, etc,.
- Published a conference paper in International ISHMT-ASTFE Heat and Mass Transfer Conference.

**Internship on CATIA** **Jun 2020 - Aug 2020**
*AP State Skill Development Corporation (APSSDC)* *Remote*

- Practical experince with Catia V5 software.
- Designed a Turbocharger assembly for automobiles using Catia.

## Education

**Bachelor of Technology in Mechanical Engineering** **2017-2021**
*SRM University, AP* *CGPA: 8.51*
Awarded with merit scholarship of 50% tuition fee.

**Board of Intermediate Education Andhra Pradesh, BIEAP ($12^{th}$)** **2015-2017**
*Narayana Jr. College* *Percentage: 95.1%*

**Central Board of Secondary Education, CBSE ($10^{th}$)** **2014-2015**
*Sri Chaitanya Techno School* *CGPA: 9.4*

## Certifications

**Algorithmic Trading & Quantitative Analysis Using Python** **Mar 2023 - Apr 2023**
*Udemy* *Certificate*

**Introduction to Programming with MATLAB** **Apr 2020 - May 2020**
*Coursera* *Certificate*

**Autodesk Fusion 360 Integrated CAD/CAM/CAE** **Feb 2020 - Mar 2020**
*Coursera* *Certificate*

**Robo Feast 2018** **Feb 2018**
*Certificate of Participation*
Participated in the 2 days National Level Workshop cum Championship on "*WiFi-Controlled Robotics*" organized by ***Revert Technology Pvt. Ltd. & EDC, IIT Roorkee.***

** Click on hyperlinks for additional information and resources